MULTIMETER TEST LEADS
From Scratch

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If you are an electronics hobbyist and need to measure low currents and low voltages, you can buy any inexpensive multimeter. However, test leads of such meters are fragile. That's what led me to make my own multimeter test leads from readily available material. These test leads cost you almost nothing as the required material, except wires, must be already available with most of you. Note that these test leads work fine for low voltages only.

Material required

1. Two empty ball pen tubes
2. Wires
3. Refill
4. Aluminium foil
5. Two steel pins
6. Insulation tape
7. Soldering iron (used to connect wires to steel pins)

The process

Plug design. Cut two 2cm long pieces from the ballpen refill so that these can be inserted into ports of the multimeter. These refill pieces act as plugs. As the refill is made of a plastic material, cover refill pieces with aluminium foil to provide conduction.

Using insulation tape, affix one end of black and red wires to the refills, respectively; these wires represent common (ground or negative) and positive probes.

Probe and tip design. Use empty pen tubes as probes, and steel pins as tips. Insert other ends of wires through the pen tubes and connect to steel pins with the help of a soldering iron. Affix the ends with the help of insulation tape.

Testing. Test the leads to ensure that these work. Now you are good to go! EFY

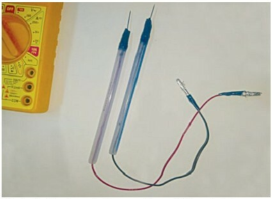

Fig. 1: Test leads made from readily available materials

Fig. 2: Material required

Fig. 3: Cut two 2cm lengths from the refill

Fig. 4: Wrap the refill pieces in aluminium foil

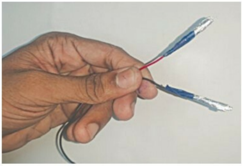

Fig. 5: Affix red and black wires to the foil-wrapped refill pieces